Swati Aggrawal

swatiagg357@gmail.com | github.com/swatiaggrawal |linkedin.com/in/swatiaggrawal

PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. Experienced in developing AI-powered applications, backend services, and machine learning solutions using Python, FastAPI, Docker, AWS, and cloud technologies. Hands-on experience with Generative AI, NLP, LLM-based applications, REST APIs, and deploying scalable AI solutions. Passionate about building practical AI systems that solve real-world business problems.

SKILLS

- **Programming:** Python, Java, SQL, C++, Bash
- **Generative AI & NLP:** Large Language Models (LLMs), Prompt Engineering, Generative AI, NLP, Sentence Transformers, Embeddings, Transformer Models
- **Backend & APIs:** FastAPI, REST APIs, API Integration, Backend Development
- **Cloud & DevOps:** AWS, Docker, Git, GitHub, Linux
- **Software Engineering:** OOP, Data Structures & Algorithms, Unit Testing, Agile/Scrum

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Computer Vision, Deep Learning and Applied AI, Natural Language Processing, Machine Learning, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed an AI-powered document intelligence platform for semantic comparison and automated change detection across document versions.
- Leveraged AI-assisted development tools to accelerate prototyping, debugging, and implementation.
- Designed RESTful endpoints using FastAPI for automated document analysis and change tracking workflows.
- Developed backend workflows for document processing and semantic comparison.
- Integrated transformer-based ML models into application pipelines for automated inference.
- Implemented semantic matching workflows to automate identification of added, removed, and modified content across document versions.
- Containerized the application using Docker and deployed it on AWS ECS Fargate.
- Configured application monitoring and logging using AWS CloudWatch.
- Experimented with prompt engineering and transformer-based NLP techniques for improving semantic document understanding.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS ECS Fargate, CloudWatch, REST APIs

PosePal Real-Time Yoga Pose Classification System

- Developed a real-time computer vision application that classifies yoga poses from live webcam input.
- Built and integrated a CNN-based image classification model for multi-class pose recognition.
- Implemented image preprocessing, normalization, and data augmentation techniques to improve model performance.
- Integrated model inference into a lightweight web application for real-time predictions.
- Applied frame-level prediction smoothing to reduce prediction fluctuations during live inference.
- Developed and integrated a browser-based user interface for real-time webcam-based pose classification.

- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

Metal Defect Detection using Computer Vision

- Developed computer vision models for automated defect detection using CNN, SVM, and Random Forest.
- Built preprocessing, training, and evaluation pipelines.
- Compared deep learning and traditional ML approaches.
- Performed model evaluation using accuracy, confusion matrices, and performance analysis.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) — Nov 2025

- Evaluated pretrained Vision Transformer architectures under Task-Incremental learning settings.
- Designed training, evaluation, and benchmarking pipelines using Python and PyTorch.
- Automated experiment tracking and performance analysis across continual learning scenarios.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

Machine Learning for Air Pollution [IEEE Publication] — May 2023

- Published IEEE conference paper evaluating machine learning approaches for environmental prediction and monitoring.
- Evaluated supervised ML algorithms including Random Forest, SVM, Decision Trees, and Neural Networks across environmental datasets.
- Performed comparative analysis using model performance metrics to assess accuracy, robustness, and generalization.
- Presented scientific findings through structured research documentation and publication.